

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
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Due Date	30/3/26		

Experiment 7: Bagging, Boosting and Stacked Ensemble Models

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1 Objective

- Implement Bagging, Boosting and Stacking models.
- Compare ensemble models.
- Analyze bias and variance.

2 Dataset

Wisconsin Breast Cancer Dataset

- Samples: 569
- Features: 30
- Classes: Benign, Malignant

3 Exploratory Data Analysis

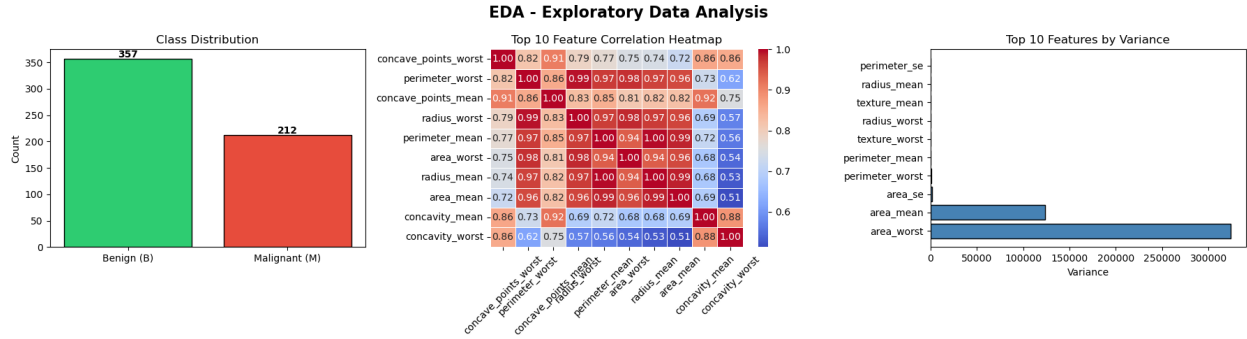


Figure 1: EDA: Class distribution, correlation heatmap and feature variance

4 Hyperparameter Results

4.1 Bagging

Table 1: Bagging Hyperparameter Evaluation

n_estimators	max_samples	Avg CV Accuracy (%)	Avg CV F1 Score
10	0.5	94.07	0.9163
10	0.8	94.51	0.9226
50	0.5	95.82	0.9434
50	0.8	96.48	0.9526
100	0.5	95.82	0.9436
100	0.8	95.60	0.9399

4.2 Boosting

Table 2: Boosting Hyperparameter Evaluation

n_estimators	learning_rate	Avg CV Accuracy (%)	Avg CV F1 Score
50	0.01	92.09	0.8863
50	0.10	95.16	0.9336
50	1.00	96.70	0.9546
100	0.01	92.53	0.8937
100	0.10	96.04	0.9455
100	1.00	96.48	0.9523
200	0.10	96.70	0.9542
200	1.00	97.14	0.9610

4.3 Stacked Ensemble

Table 3: Stacked Ensemble Evaluation

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score
SVM + NB + DT	Logistic Regression	97.36	0.9638
SVM + DT	Logistic Regression	97.36	0.9642
NB + DT	Logistic Regression	94.51	0.9244

5 Performance Comparison

Table 4: Performance Comparison of Ensemble Models

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	97.37	1.000	0.9286	0.963
Boosting	97.37	1.000	0.9286	0.963
Stacked Ensemble	97.37	1.000	0.9286	0.963

6 Confusion Matrix

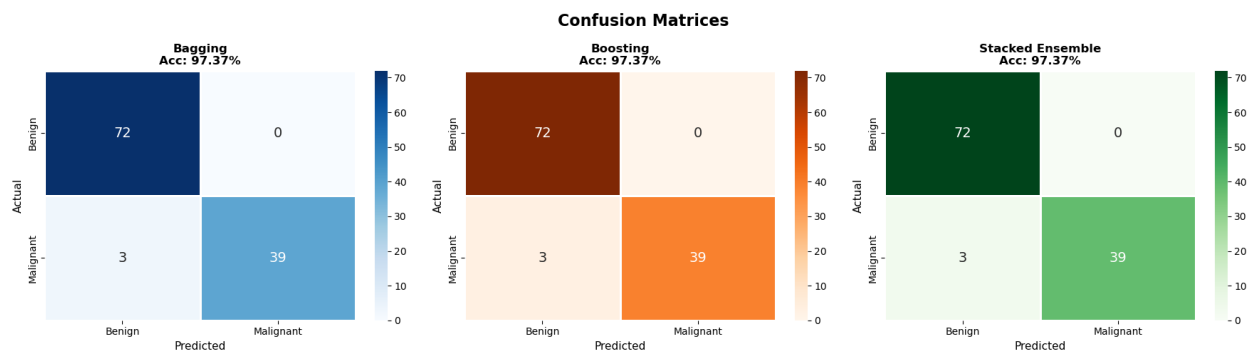


Figure 2: Confusion Matrices of Ensemble Models

7 ROC Curve

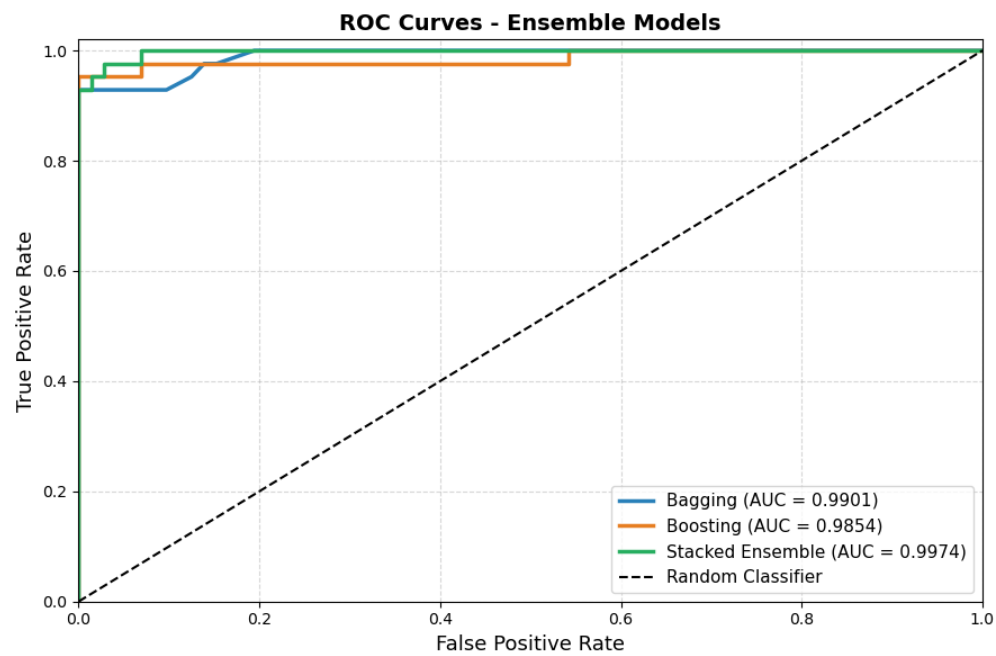


Figure 3: ROC Curve Comparison

8 Performance Graph

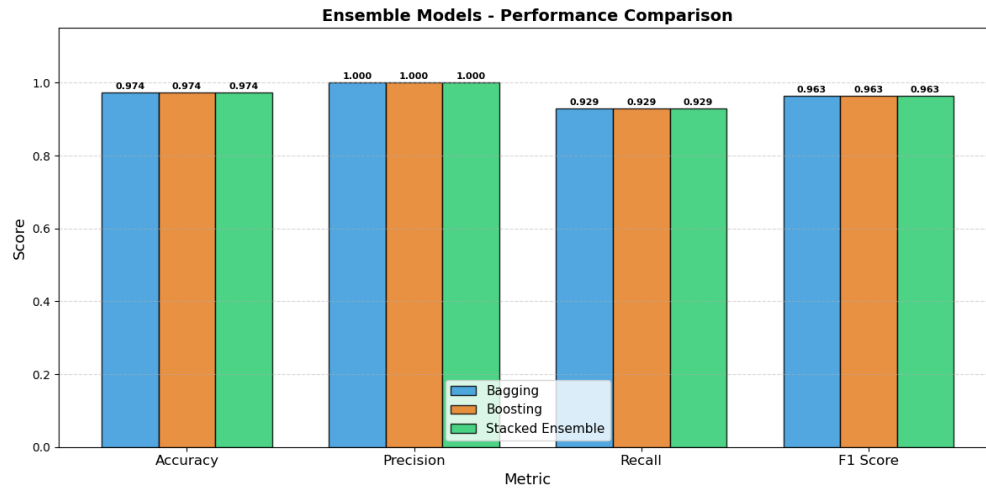


Figure 4: Performance Comparison of Ensemble Models

9 Observations

- Bagging reduces variance and improves stability.
- Boosting improves model accuracy by reducing bias.
- Stacking combines multiple models for better performance.
- All ensemble methods achieved high accuracy.

10 Conclusion

Bagging, Boosting and Stacked Ensemble models were implemented and evaluated. Ensemble techniques improved classification performance and generalization compared to individual models.